Q-Vision — POC Prerequisites & Setup Guide

> Reference document for prototype testing with iPhone camera setup. Updated: 9 March 2026

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1. Materials Required

Item	Specification	Purpose
Sample material — 6mm	Crushed stone aggregate, 6mm grade	Testing 6mm classification
Sample material — 12mm	Crushed stone aggregate, 12mm grade	Testing 10mm class (8–12mm range)
Sample material — 20mm	Crushed stone aggregate, 20mm grade	Testing 20mm classification
Steel ruler (30cm)	Standard steel ruler with mm markings	One-time px/mm calibration
Dark surface	Dark cloth / dark cardboard / dark tray	High contrast background for segmentation
iPhone	Any model with auto-focus camera	Fixed camera for POC
Phone mount	Tripod / clamp arm / fixed stand	Consistent height for all photos

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2. Camera Setup

Position

- Mount iPhone **pointing straight down** (perpendicular to the surface)
- Fix at a consistent height (recommended: **40–60 cm** for tabletop POC)
- Use default **4:3 photo mode** — no special camera settings needed
- Any aspect ratio works (4:3, 16:9, 1:1)

Focus

- **Tap on the stones** on the iPhone screen before each photo to lock focus
- Stone edges must be **sharp and clearly defined** — blurry edges = bad segmentation
- Individual stone textures should be visible

Lighting

- Use **indirect daylight** or a **diffused lamp**
- Avoid harsh direct light that creates strong shadows
- Shadows between stones confuse the segmentation algorithm
- Keep lighting consistent across all test photos

Framing

- Fill **at least 70–80%** of the frame with stones
- Extra background space is acceptable — pipeline filters it out
- Avoid having most of the frame empty (< 40% fill = bad results)

```
Ideal ✅                    Acceptable ⚠              Bad ❌

┌──────────────┐           ┌──────────────┐          ┌──────────────┐
│ stones stones│           │              │          │              │
│ stones stones│           │  stones      │          │              │
│ stones stones│           │  stones      │          │    stones    │
│ stones stones│           │              │          │              │
└──────────────┘           └──────────────┘          └──────────────┘
 ~80-100% filled           ~40% filled               ~10% filled
```

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3. One-Time Calibration (px/mm)

Setup

```
       iPhone (looking straight down)
           📱
            │
            │   (fixed height)
            │
            ▼
     ┌──────────────────┐
     │  stones stones    │
     │   stones  📏      │  ← ruler flat ON the stones / same surface level
     │  stones stones    │
     └──────────────────┘
        dark surface
```

Steps

1. Spread sample stones on the dark surface
2. Place the steel ruler **flat on top of the stones** (same level as stone surfaces)
3. Take ONE photo from the fixed iPhone position
4. Open the photo on Mac (Preview or any editor)
5. Zoom into the ruler
6. Count pixels between two marks **far apart** (e.g. 0cm to 10cm = 100mm)
7. Calculate: `px_per_mm = pixel_count ÷ mm_distance`
8. Set value in `config.py`: `FIXED_PX_PER_MM = <your_value>`
9. **Remove the ruler — never needed again** (as long as camera height stays the same)

Example

- Ruler: 0cm to 10cm = 100mm
- Pixel distance between those two marks = 520 pixels
- `px_per_mm = 520 / 100 = 5.2`
- Set `FIXED_PX_PER_MM = 5.2`

Important

- Ruler must be at **same height/level** as the stones (not on the camera, not on the ground below)
- Use marks **far apart** for accuracy (100mm+ recommended)
- If you change camera height, recalibrate

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4. Software Prerequisites

Dependencies

```
# Activate virtual environment
source venv/bin/activate

# Install dependencies
pip install -r requirements.txt
```

Pre-test Code Fixes Required

- [ ] **EXIF rotation fix** — iPhone JPGs have EXIF orientation tags; `cv2.imread` ignores them → photos appear rotated. Fix needed in `utils.py`
- [ ] **Dashboard histogram bug** — Uses hardcoded `5.0` px/mm instead of actual calibrated value. Fix in `dashboard.py`
- [ ] **Set `FIXED_PX_PER_MM`** in `config.py` after calibration photo

Run Tests

```
python -m pytest tests/ -v
```

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5. Testing Protocol

Per Sample Material

1. Spread ONE material type on the dark surface (don't mix)
2. Take photo from fixed iPhone position
3. Save to `sample_images/` folder with descriptive name

CLI Test Commands

```
# 6mm sample
python main.py sample_images/6mm_sample.jpg --truck-id POC-001 --expected
6mm

# 12mm sample (maps to 10mm class: 8–12mm range)
python main.py sample_images/12mm_sample.jpg --truck-id POC-002 --expected
10mm

# 20mm sample
python main.py sample_images/20mm_sample.jpg --truck-id POC-003 --expected
20mm
```

Dashboard Test

```
streamlit run dashboard.py
# Open http://localhost:8501
# Upload photo → select expected material → Run Analysis
```

What to Check

- [ ] Classification matches the actual material
- [ ] Confidence is reasonably high (> 60%)
- [ ] Histogram shows particles concentrated in the correct size range
- [ ] No "MIXED LOAD" warning for single-material samples
- [ ] Zone breakdown shows consistency across all 5 zones

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6. Classification Reference

Material	Size Range	Class Label	Threshold
6mm	4 – 8 mm	`6mm`	≥ 80% of particles
12mm	8 – 12 mm	`10mm`	≥ 80% of particles
20mm	16 – 25 mm	`20mm`	≥ 80% of particles
Mixed	—	`MIXED LOAD`	No class dominates

> **Note:** 12mm material falls within the `10mm` class (8–12mm range). Particles between 12–16mm fall into "other".

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7. Troubleshooting

Problem	Likely Cause	Fix
Photo appears sideways	EXIF rotation not handled	Apply EXIF fix in `utils.py`

Problem	Likely Cause	Fix
All particles classified as wrong size	Wrong `px_per_mm` value	Recalibrate with ruler
Too many particles detected (noise)	Background not dark enough	Use darker surface
Too few particles detected	Stones blend into background	Improve lighting/contrast
"MIXED LOAD" on single material	Segmentation picking up noise	Tune `MIN_CONTOUR_AREA` in `config.py`
Blurry segmentation results	Camera not focused	Tap to focus on stones before shooting

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8. Learnings Log

> Track observations, parameter adjustments, and results from each test session below.

Date	Test	Observation	Action Taken